

Not In Our Time: Transport-Based Content Selection

An Inspectable, Manipulable Alternative to RAG — Demonstrated on Literary Discussion Podcasts for 15 Victorian Novels

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“Fog everywhere. Fog up the river, where it flows among green aits and meadows; fog down the river, where it rolls defiled among the tiers of shipping and the waterside pollutions of a great (and dirty) city. Fog on the Essex marshes, fog on the Kentish heights.”

— Charles Dickens, *Bleak House* (1853), Chapter 1

The Question

A reader who surveys a novel broadly and a reader who studies it narrowly will assemble different evidence for discussion. Do they arrive at different discussions?

Google’s NotebookLM popularised LLM-generated podcast conversations, but the conversational design behind such systems is largely unexamined. What makes a multi-voice discussion *cohere*—sound like a genuine conversation rather than parallel monologues? What levers does a designer have, and which ones actually matter?

We implement two reading strategies as passage-selection algorithms applied to **15 Victorian novels**, then generate structured multi-voice literary discussions from each selection.

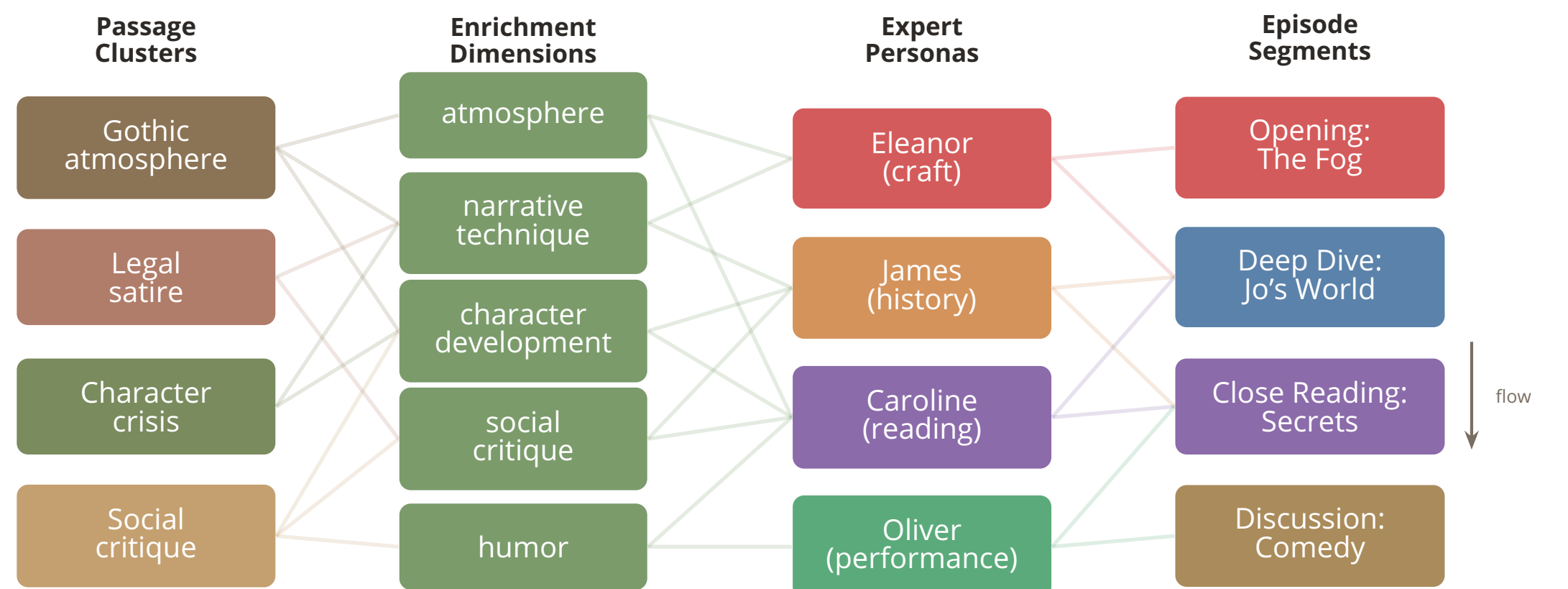


Seven authors (Dickens, Eliot, Gaskell, Forster, Collins, Gissing, Oliphant), seven decades (1850–1924).

“And all that night the coffin stands ready by the old portmanteau; and the lonely figure on the bed, whose path in life has lain through five and forty years, lies there with no more track behind him that any one can trace than a deserted infant.”
— Bleak House, Ch. 11 — The death of Nemo

Min-Cost Flow Selection

Passages flow through enrichment dimensions and expert demand nodes to episode segments. Every arc is inspectable and adjustable.



Two Reading Strategies
Synoptic (min-cost flow): 59 passages from 28 chapters. Favours character development (+8.3 pp) and plot (+10.2 pp). Zero LLM tokens, <1 s.
Analytical (embeddings + LLM curation): 32 passages from 20 chapters. Favours social critique (+13.4 pp) and thematic depth (+6.9 pp).
No-passages: LLM discusses from prior knowledge alone.

Why Not RAG?

Standard RAG retrieves by embedding similarity and feeds results to an LLM. Effective but opaque: the user cannot see *why* a passage was selected, cannot adjust criteria, and cannot explore how different priorities reshape the output.

Min-cost flow makes every decision visible as a **cost**, a **flow**, and a **constraint**. Solving a flow costs <1 s and zero LLM tokens—a **thinking tool** for rapid exploration of alternatives.

References

Fish, S. (1980). *Is There a Text in This Class?* Harvard UP.
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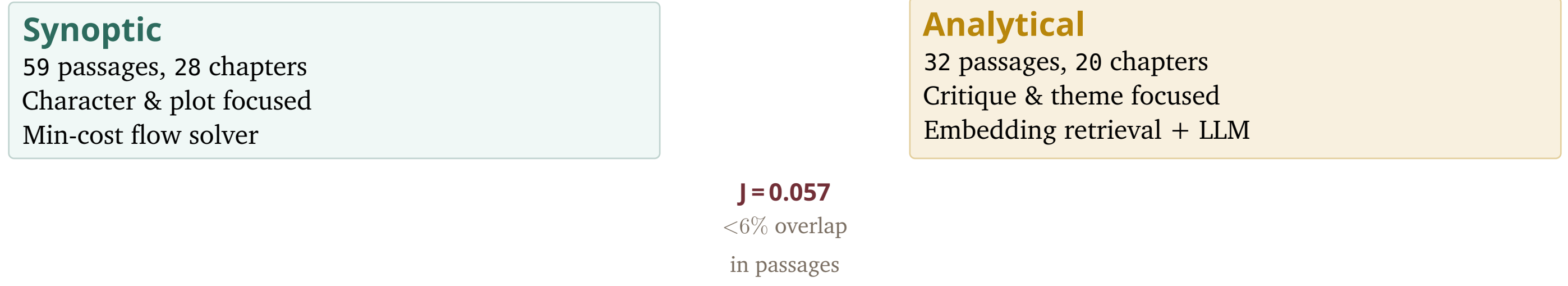
Interactive Demo & Acknowledgements

bleakhouse-demo.fly.dev Browse 180 conditions, listen to episodes, inspect passage backlinks.
Thanks to the Ohio State University College of Engineering and LexisNexis Legal & Professional.

The Reader Makes the Reading

The pipeline privileges expert identity at every stage — enrichment dimensions match demand profiles, the selector optimises for demand satisfaction, host preparation targets specific experts, and the script prompt reinforces perspectives. Convergence follows from this engineering.

The system exposes four independently manipulable levers: **expert personas** (demand profiles that control what each speaker seeks); **passage grounding** (selected passages that anchor quotation and prevent confabulation); **conversational design** (prompt-level specifications for quote handling, reactive language, and turn structure); and **segment continuity** (bridging between independently generated segments).



Yet: Expert Airtime Stable to ±2%
Eleanor Hartley (literary critic): **32–37%** James Blackstone (social historian): **30–37%** Caroline Woodcourt (close reader): **29–34%**
Each stage reinforces the framework. The convergence is suggestive of Fish’s (1980) thesis — but the engineering is the mechanism, and we name it honestly.

Two Panels Discuss *Bleak House*

Literary Panel

Eleanor Hartley (Literary Critic):
“What strikes me about that Nemo sentence is the grammatical audacity of it. ‘A deserted infant’—Dickens reaches for the most helpless possible image of erasure. The infant image should be about beginning, and Dickens flips it into a closing.”

James Blackstone (Social Historian):
“The fog is not decoration. It is the Court of Chancery made visible. ‘Lies there with no more track behind him that any one can trace than a deserted infant.’ That sentence is the whole novel compressed into a single image.”

Caroline Woodcourt (Close Reader):
“Readers sometimes use the fog as a kind of permission—‘this is a great symbolic novel, I’m in the presence of Literature with a capital L’—and then they stop actually listening to what Dickens is telling them. What moves me is something far quieter—the moment where a human life is described as leaving no trace.”

Interdisciplinary Panel

Sarah Chen (NLP Scientist):
“That is a noun phrase with four pre-nominal modifiers, each compounded—and then ‘brazen-faced’ pulls the floor out. The first three feel like nautical specifications—they’re doing the register of technical competence—and the fourth is an insult.”

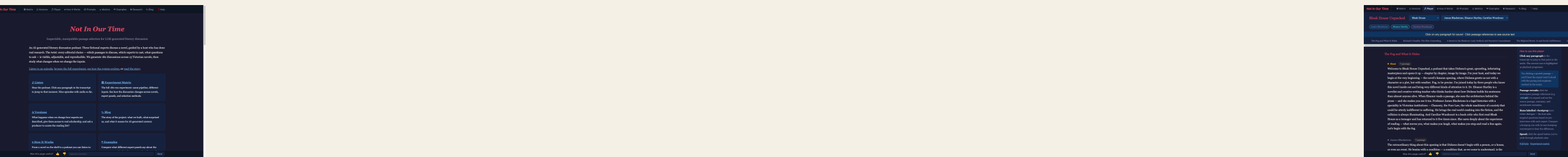
Elena Volkov (Musicologist):
“Institutions don’t announce their corruption—they announce their solidity, their procedure, their continuity. I spent years studying how Soviet conservatories presented themselves as nurturing talent while systematically destroying composers who stepped out of line.”

Rebecca Martinez (Astronomer):
“The ships of Law and Equity aren’t sunk, they’re in reserve. In my field, we’d say the system is in a quasi-stable state. The energy is there. Nothing has been resolved. The fog is the natural atmosphere of a system that has never actually resolved anything.”

The experts bring their **metaphors** (quasi-stable states, leitmotifs, phrase structure) but not their **methods** — the scaffold imports literary-critical norms regardless of the persona.

Deliberate Stereotypes

The expert personas are deliberately *caricatures*: the literary critic always finds prose rhythm; the social historian always finds institutional critique; the close reader always finds texture. By making the stereotypes explicit—encoded as **demand vectors** rather than implicit in a prompt—the system turns editorial bias into a first-class object that can be inspected, compared, and argued about.

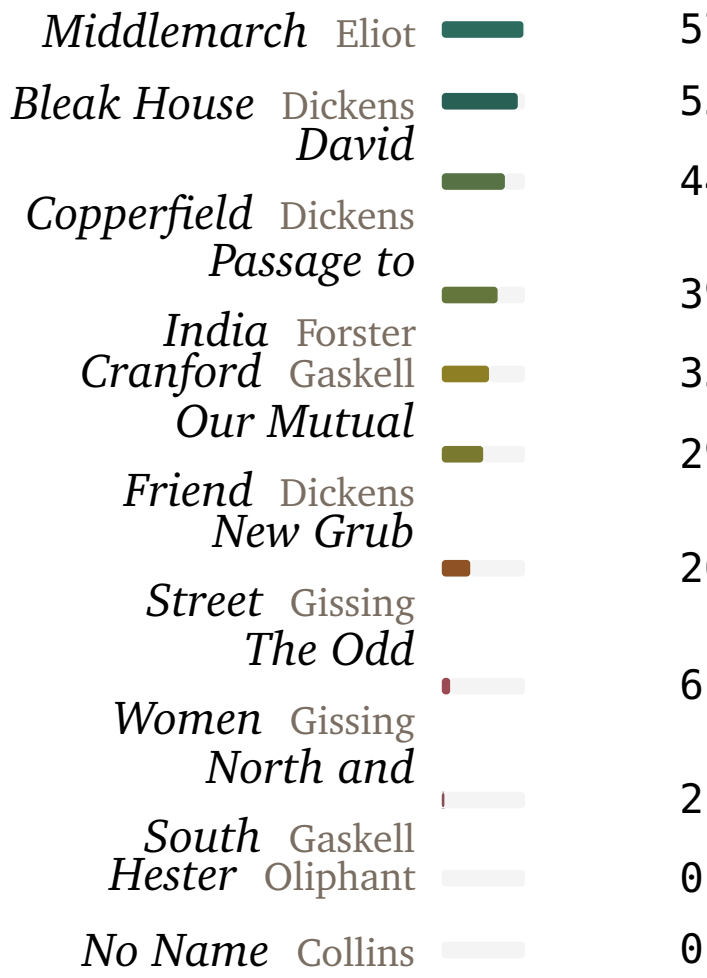


The Knowledge Gradient

Without passage grounding, the LLM produces graduated pastiche — not hallucination. Fidelity tracks novel prominence.

82% of invented quotes draw 90–100% of their words from the novel’s own vocabulary.
Mean overlap: 95.1%.

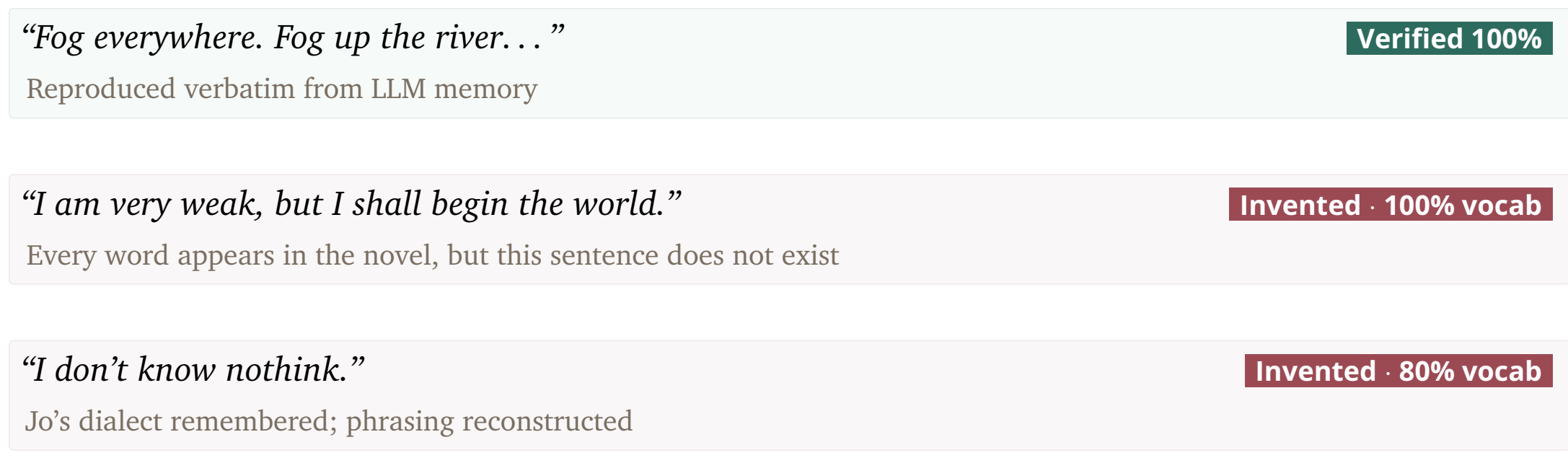
Ungrounded Quote Verification (%)



Grounded conditions achieve **97–98%** for *all* novels. The gradient reflects analytical reinforcement in training data (Carlini et al., 2021, 2023), not prose style.

Pastiche in Action

“Name, Jo. Nothing else that he knows on. Don’t know that everybody has two names. Never heerd of sich a think. No father, no mother, no friends. Never been to school.”
— Bleak House, Ch. 11 — Jo’s testimony



Key Findings

- Framework dominates strategy.** Two reading algorithms select <6% overlapping passages yet produce discussions with identical expert airtime (±2%). This convergence follows from a pipeline that reinforces expert identity at every stage — suggestive of Fish’s thesis, but the engineering is the mechanism.
- Pastiche, not hallucination.** Ungrounded LLMs write *in the style of* a novel, not *from* it. Fidelity tracks training-data exposure: a measurable knowledge gradient from *Middlemarch* (57%) to *Hester* (0%).
- Scaffolding is content-independent.** Host preparation raises questions from ~1 to ~11 per segment, uniformly across all 15 novels (5.3× to 14.1×). Dynamics are designed, not emergent.

Framework > Grounding > Strategy
The decomposition has a clear hierarchy. Framework determines character, grounding gates accuracy, strategy is surprisingly irrelevant.